

Krish Shah

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SUMMARY

Data Science Master's candidate (UW-Madison) adept at leveraging machine learning and statistical analysis to drive business value. Experience includes developing recommendation systems (60% CTR increase), time-series forecasting (40% sales boost), and NLP pipelines (15% response time improvement). Proficient in Python, SQL, TensorFlow, PyTorch, and major cloud platforms. Seeking a data science internship to contribute to data-driven decision-making.

EDUCATION

University of Wisconsin-Madison , Master of Science in Data Science GPA: 3.83/4.00	August 2024 - May 2026
Sardar Patel Institute of Technology , Bachelor of Computer Engineering GPA: 8.92/10.00	August 2020 - May 2024

SKILLS SUMMARY

Languages: Python, R, SQL, Java, Scala
Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Spark MLlib, Transformers, XGBoost, spaCy, NLTK
Analysis: NumPy, Pandas, Matplotlib, Seaborn, Plotly, Tableau, QlikView, Power BI, Looker, Hypothesis Testing
Techniques: Statistical Modeling, Predictive Analytics, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Time Series Analysis, Recommender Systems, Anomaly Detection, Data Mining
Databases: MySQL, PostgreSQL, NoSQL, Snowflake, MongoDB, Cassandra, Data Warehousing, Lake, ETL, ELT
Cloud: AWS (SageMaker, EC2, S3, Glue, Athena, Redshift, Lambda), Azure (Databricks, Blob Storage, Synapse, Factory), GCP (Vertex AI, Compute Engine, BigQuery, Dataflow), Big Data (Hadoop, Kafka, Spark, Flink)

EXPERIENCE

- Mahavir Travels** Remote
Data Scientist January 2024 - August 2024
 - Engineered** a time-series model (ARIMA) using SQL, **boosting sales by 40%** with a **90% accuracy**.
 - Designed and implemented** a collaborative filtering recommendation system using Databricks, PySpark, and MLlib, **increasing click-through rates by 60%** and improving customer satisfaction by **90%**.
 - Spearheaded** an anomaly detection system (Isolation Forest) to mitigate **95%** of fraudulent bookings and reduce operational costs by **25%**, integrated with Kafka and Apache Airflow for pipeline automation.
- Great Lakes Institute of Management** Chennai, India
Deep Learning Research Intern June 2023 - August 2023
 - Constructed** a real-time video captioning system using Transformers and R-CNN with PyTorch and spaCy for NLP, enhancing video analysis capabilities.
 - Optimized** an OpenCV frame extraction algorithm with Savitzky-Golay filtering, **reducing processing time by 40%** and memory usage by **55%** for real-time applications in sports commentary.
 - Examined** in-store consumer behavior using Computer Vision and applied the Apriori algorithm (mlxtend) for product association analysis, to provide insights for increasing store and product revenue and improving market understanding.
- Indian Institute of Technology Patna** Patna, India
Deep Learning Research Intern January 2023 - June 2023
 - Achieved 99.68% accuracy** in brain tumor detection/classification using TensorFlow (VGG19, MobileNet) with transfer learning, preprocessing MRI scans with gamma correction and CLAHE via scikit-image.
 - Optimized** the model based on radiologist feedback from IGIMSS Cancer Hospital, expanding the dataset and fine-tuning for generalization using a PostgreSQL-based Data Lake architecture, improving diagnostic accuracy.

PROJECTS

- Sentiment Analysis Pipeline with AWS (SageMaker, Glue, Athena, S3, CloudWatch, Data Wrangler)** Created an AWS-based machine learning pipeline for sentiment analysis on 23,000+ customer reviews, **improving customer service response time by 15%** with BERT fine-tuning.
- Network Intrusion Detection System (NIDS) (Python, TensorFlow, Databricks, Splunk, MongoDB)** Implemented a Python-based NIDS, achieving a **97% detection rate** using anomaly detection and Databricks.
- Restaurant Data Analysis and Recommendation System (Python, R, Power BI, SQL, Snowflake)** Processed 10,000+ restaurant records, building a recommendation system, **boosting restaurant ratings from 3.6 to 4.1**.
- Generative AI-Powered Chatbot for PDFs (Python, Langchain, Google Gemini Pro, Streamlit)** Created a PDF answering chatbot with Langchain and Gemini, enhancing response accuracy using FAISS semantic and spaCy for NLP.

PUBLICATIONS

- Computer-aided diagnosis for multi-class classification of brain tumors using CNN features via transfer-learning** Published in Multimedia Tools and Applications, Springer Journal US. DOI: [10.1007/s11042-025-20751-z](https://doi.org/10.1007/s11042-025-20751-z)